TAX SAHAYAK

AI-BASED FINANCIAL ASSISTANT

PYTHON PROGRAMMING LAB

SUBMITTED BY:

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PROBLEM STATEMENT

In real life, many users face difficulties in:

- Calculating EMI manually using formulas
- Understanding tax calculations and rules
- Finding reliable answers to financial questions quickly

Existing solutions are often:

- Complicated
- Time-consuming
- Not interactive

OUR SOLUTION

- Automates EMI & Tax Calculations
- Users just enter basic inputs (loan amount, income, etc.), and the system instantly computes accurate results using backend logic.
- Provides Real-Time Results
- Calculations are processed instantly through the FastAPI backend, eliminating manual effort and delays.
- Integrates AI Chatbot for Queries
- Users can ask financial questions (e.g., tax rules, loan advice), and the AI generates intelligent, context-aware responses.
- Overall: A unified platform that combines calculation tools and AI assistance for better financial decision-making.

IMPLEMENTATION (HOW IT WORKS)

STEP 1: USER INTERACTION

User enters :
- Loan details (EMI)
- Income (Tax)
- Or asks a question in chatbot

STEP 2: BACKEND PROCESSING (Python)

Python backend (FastAPI):
- Processes user inputs
- Calculates EMI and tax
- Sends chatbot query to Groq API

Step 3: AI Processing

Groq API:

- Understands user query
- Generates financial response

Step 4: Output

User gets:

- EMI result
- Tax calculation
- Chatbot response

FLOW:

USER

↓

REACT UI

↓

PYTHON BACKEND

↓

AI/DATA

↓

RESULT

Tech Stack

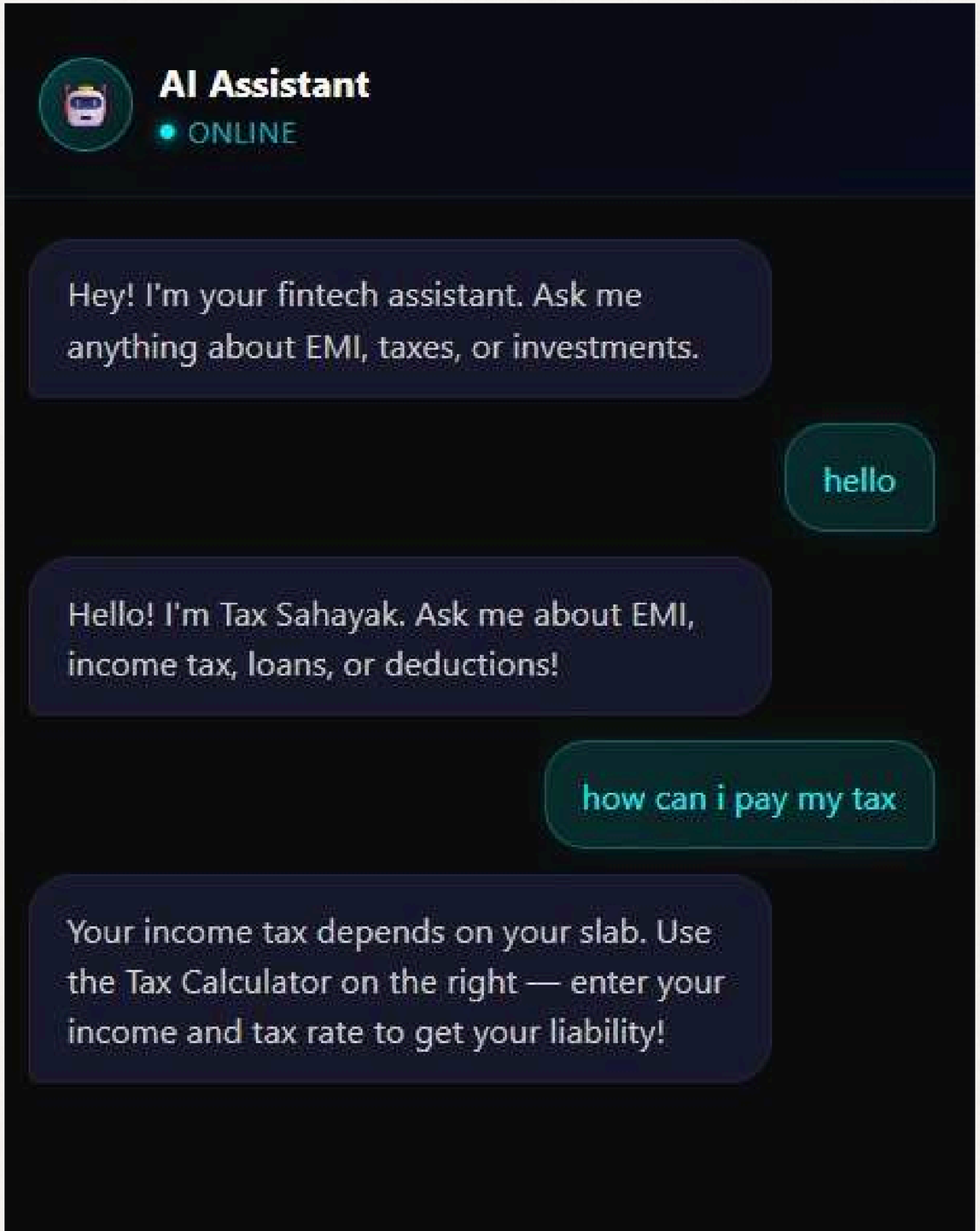
Technologies Used

1. **Frontend:**
- React.js for dynamic UI
- Tailwind CSS for styling

2. **Backend:**
- Python for logic implementation
- FastAPI for building APIs

3. **AI Integration:**
- Groq API for chatbot responses

4. **Tools:**
- GitHub for version control and project management



1.AI Fintech Assistant (Tax Sahayak)

2.AI chatbot for EMI, taxes, loans, and deductions

3.Provides instant financial guidance through chat

4.Helps calculate tax based on income slab

5.Simple, fast, and user-friendly

6.Example:
User: “How can I pay my tax?”
Assistant: Suggests using a tax calculator and explains liability

Fintech Dashboard

Smart financial calculators at your fingertips

4 May 2026

EMI Calculator

PRINCIPAL AMOUNT (₹)

575

ANNUAL INTEREST RATE (%)

2.5

TENURE (MONTHS)

12

Calculate EMI

MONTHLY EMI

₹48.57

TOTAL INTEREST	TOTAL PAYMENT
₹7.82	₹582.82

Tax Calculator

ANNUAL INCOME (₹)

4546

TAX RATE (%)

20

Calculate Tax

TAX LIABILITY

₹909.20

NET TAKE-HOME INCOME

₹3,636.80

Features and Benefits

Key Features

- AI chatbot for answering finance-related questions
- EMI calculator for loan planning
- Tax calculator for income-based estimation
- Real-time processing and instant results
- Clean and modern user interface

Benefits

- Saves time by eliminating manual calculations
- Provides quick and accurate financial results
- Easy-to-use interface, suitable for non-experts
- High-speed processing using FastAPI backend

TAX SAHAYAK – SYSTEM ARCHITECTURE FLOWCHART

User → Frontend → Backend → AI/Logic → Output

USER
Enters details /
Asks a question

FRONTEND
React.js + Tailwind CSS
(Handles UI and User Interaction)

- User inputs loan details / income / query
- Displays results in a modern UI
- Sends requests to backend APIs

BACKEND
Python FastAPI
(Handles Routing, Processing & Logic)

- Receives request from frontend
- Validates and processes the data
- Performs calculations or forwards query to AI
- Sends response back to frontend

EMI / TAX LOGIC
Performs Calculations
(EMI Calculation / Tax Calculation)

GROQ API (AI LAYER)
Processes Natural Language Query
and Generates Response

- Understands user query
- Generates intelligent financial responses
- Returns result to backend

RESPONSE PROCESSING
Combines and formats the result
(Calculation Result / AI Response)

OUTPUT TO USER
Displays EMI Result / Tax Result /
Chatbot Response

- Instant and accurate results
- User-friendly display
- Enhances user experience

TECHNOLOGY STACK

Frontend	: React.js + Tailwind CSS
Backend	: Python FastAPI
AI Layer	: Groq API
Tools	: GitHub

KEY FEATURES

- AI Chatbot for Finance Queries
- EMI Calculator
- Tax Calculator
- Real-time Processing
- Modern & Responsive UI
- Easy to Use

CONCLUSION

Tax Sahayak is a smart and efficient financial assistant that integrates:

- Python backend for processing
- AI capabilities for intelligent responses

It helps users by:

- Reducing manual calculations
- Providing instant financial insights
- Enhancing user experience